The inventory gerbe couples on the torus

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Abstract

A supply chain, abstracted to a small category $\mathcal{C}$, carries fungible inventory as local stock-data glued along logistics operations up to a relabelling group $K = \mathrm{Aut}F$ of the stock fibre F; the gluing datum is an $\mathrm{Aut}F$-gerbe over the nerve $N(\mathcal{C})$, and following Giraud its obstruction to a strict global inventory is *staged*: a nonabelian band $\beta \in H^1(N(\mathcal{C}); \mathrm{Out}K)$ must vanish before an abelian class $[\mathfrak{g}] \in H^2(N(\mathcal{C}); Z(K))$ is even defined. We ask whether the stages are independent. Over a one-dimensional base they are, but over a two-dimensional one they need not be. We prove a general *δ-commutator lemma*: for an abelian-kernel extension of coefficient groups and a band $\rho\colon \mathbb{Z}^2 \to Q$ on the torus, the connecting map is given exactly by the commutator of any two lifts, $\delta(\rho) = [[A, B]]$, in the coinvariants $H^2(T^2; M)$. Consequently a structure group couples on T^2 – a nonzero band forcing a nonzero degree-two class – if and only if some commuting outer pair admits no commuting lift, which forces $\mathrm{Aut}K \to \mathrm{Out}K$ to be non-split. We then locate the phenomenon precisely: for the n-lot ring $K = D_n$ the inventory class is essentially nonzero exactly when $4 \mid n$, and across an enumerated family of order-16 nilpotency-class-two groups only $D_4 \times C_2$ and $Q_8 \times C_2$ couple. Coupling is a rare, identifiable feature of gauge-structured stock, not of mere interchangeability.

1 Introduction

The problem. A supply chain is a small category $\mathcal{C}$: objects are nodes (warehouses, plants, ledgers), morphisms are logistics operations, composition is sequencing. Inventory is a datum glued along the morphisms. When two local inventory charts are comparable only *up to a relabelling* of stock states – that is, up to the automorphism group $K := \mathrm{Aut}F$ of the stock fibre F – the gluing datum is no longer a presheaf but an $\mathrm{Aut}F$-gerbe over the nerve $N(\mathcal{C})$. A *strict global inventory* is a global section: a single coherent choice of trivialization valid across the whole chain. The question with economic teeth is: *when does one exist?* A failure is a compositional defect – stock-keeping that cannot be reconciled globally no matter how the local books are kept.

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The staged obstruction. In two companion notes we showed that the obstruction is not a single cohomology class but a *tower*, exactly matching Giraud's three-stratum classification of nonabelian gerbes [4, 1]. The relabelling group $K = \mathrm{Aut}F$ sits in two abelian-kernel extensions that share their middle term $\mathrm{Inn}K$:

$$\text{(II)}\quad 1 \to Z(K) \to K \to \mathrm{Inn}K \to 1, \qquad \text{(I)}\quad 1 \to \mathrm{Inn}K \to \mathrm{Aut}K \xrightarrow{q} \mathrm{Out}K \to 1. \tag{1}$$

The *band* is the outer transport $\beta = [q\circ\alpha] \in H^1(N(\mathcal{C}); \mathrm{Out}K)$ read off extension (I); it is a genuinely nonabelian datum, and it must vanish before the abelian *inventory class* $[\mathfrak{g}] \in H^2(N(\mathcal{C}); Z(K))$ – the stratum-three class of extension (II) – is even well-defined. A strict global inventory exists iff *both* vanish (Corollary 5, the *floor*).

The gap. The floor leaves a question the companion notes flagged but did not answer: are the two stages *independent*, or can one force the other? On a base of cohomological dimension one – e.g. a single reversible operation, $\mathcal{C} = B\mathbb{Z}$, $N(\mathcal{C}) \simeq S^1$ – they are forced apart, because $H^2(S^1; -) = 0$ annihilates every degree-two class; every band is strictifiable and the stages stack freely. Among the *free* bases $B(\mathbb{Z}^n)$ the first that can hold a degree-two class is the torus $T^2 \simeq N(B\mathbb{Z}^2)$: two commuting reversible operations. (Over a *torsion* base, coupling can already occur at rank one; see §7.) There the question is open on its face.

The result. We close it with a single computation, valid for every abelian-kernel extension of coefficient groups. Write the band as a homomorphism $\rho\colon \mathbb{Z}^2 \to Q$ into the relevant quotient ($Q = \mathrm{Out}K$ or $Q = \mathrm{Inn}K$), and let M be the (abelian) kernel. Because $\mathbb{Z}^2$ is abelian, a band is exactly a *commuting pair* of classes $\rho(s), \rho(t) \in Q$.

Theorem 1 (δ-commutator lemma; see Theorem 8)**.** *Let $1 \to M \to E \xrightarrow{\pi} Q \to 1$ be an extension of groups with M abelian, and let $\rho\colon \mathbb{Z}^2 \to Q$ be a band with lifts $A, B \in E$ of $\rho(s), \rho(t)$. Then $w := [A, B] = ABA^{-1}B^{-1}$ lies in M, and under the canonical identification $H^2(T^2; M) = M_{\mathbb{Z}^2}$ with the coinvariants, the connecting map is*

$$\delta(\rho) = [\,w\,] \in M_{\mathbb{Z}^2},$$

independently of the lifts. In particular $\delta(\rho) = 0$ iff the band admits commuting lifts, iff ρ lifts to a homomorphism $\mathbb{Z}^2 \to E$.

The method is deliberately elementary – a Schreier-style computation in the discrete, nonabelian register, requiring no gerbe machinery beyond the classical dictionary between H^2 and extensions [2, 3]. The one non-classical step is an explicit comparison chain map between the Koszul and bar resolutions of $\mathbb{Z}$ over $\mathbb{Z}[\mathbb{Z}^2]$, sending the fundamental 2-cell to the antisymmetric expression $\varphi_2(e_{st}) = [s\,|\,t] - [t\,|\,s]$; this turns the slogan "evaluate the obstruction cocycle on the torus relator $sts^{-1}t^{-1}$" into the identity $\delta(\rho) = [f(s,t) - f(t,s)]$ and pins the class to the commutator.

Consequences. Theorem 1 gives an immediately checkable *coupling criterion* (Corollary 10): a structure group K couples on T^2 – a nonzero band forcing a nonzero degree-two class – iff some commuting pair in the quotient admits no commuting lift, and this forces the governing extension to be *non-split*. Two sharp locations follow.

- *Sharpness* (Theorem 12). For the n-lot ring $K = \mathrm{Aut}(C_n) = D_n$ the inventory class $[\mathfrak{g}]$ is essentially nonzero iff the central extension (II) is non-split iff $4 \mid n$. The six-lot ring $D_6 \cong \mathbb{Z}/2 \times S_3$ – nonabelian with nontrivial centre, yet split – is the witness that "nonabelian with nontrivial centre" is necessary but not sufficient.

- *Rarity* (§6). Across an enumerated battery of order-16 nilpotency-class-two groups, only $D_4 \times C_2$ and $Q_8 \times C_2$ couple. Non-splitting alone is genuinely weaker than coupling: M_{16} and $C_4 \rtimes C_4$ need not split yet carry no coupled band.

Compositional failure of global stock-keeping is therefore a feature of a thin, identifiable slice of structure groups – gauge-structured stock with non-split automorphism tower – and is invisible to fungibility.

Status and provenance. This is a consolidation at honest grades, assembled from four proof files; the only new theorem is the general δ-commutator lemma. We record the grade of each ingredient ("lean-verified" means a finite core is machine-checked in Lean 4/Mathlib).

Result	Location	Grade	Lean
Floor: strict $\iff \beta = 0 \wedge [\mathfrak{g}] = 0$	Cor. 5	proved	—
δ-commutator lemma; coupling criterion	Thm. 8, Cor. 10	proved	—
Coupling witness $D_4 \times C_2$ on T^2	§4	computed	finite core
Sharp line: $[\mathfrak{g}] \neq 0 \iff 4 \mid n$	Thm. 12	proved	finite core
Rarity over the order-16 battery	§6	computed	—

Outline. Section 2 fixes the inventory gerbe, its two extensions, and the floor. Section 3 proves the δ-commutator lemma and the coupling criterion. Section 4 gives the $D_4 \times C_2$ witness on T^2. Section 5 proves the sharp $4 \mid n$ line. Section 6 reports the rarity enumeration. Section 7 draws the consequences for compositional correctness.

2 The inventory gerbe and its staging

The gerbe. Let $\mathcal{C}$ be a small category and $N(\mathcal{C})$ its nerve. An *inventory fibre* F is the local type of a stock-state at a node; its automorphism group $K := \mathrm{Aut}F$ is the *relabelling group* of fungible lots. When local inventory charts are comparable only up to K, the gluing data form an $\mathrm{Aut}F$-gerbe over $N(\mathcal{C})$: a transport $\alpha\colon \mathrm{Mor}\mathcal{C} \to \mathrm{Aut}K$ (re-trivializing a chart conjugates the next transport), functorial only *up to inner automorphism* of K – the discrepancy recorded by a *comparison 2-cell datum* $k_{g,f} \in K$ with $\alpha_{gf} = \mathrm{conj}(k_{g,f}) \cdot \alpha_g \alpha_f$, which we carry as part of the gerbe structure throughout (and retain in Definition 4) – together with higher coherence valued in $Z(K)$. We do *not* assume $k_{g,f} \equiv e$: the comparison datum is genuine here, and it is precisely its retention that separates the honest floor (D2) from the naive variant (D1) discussed in Remark 6. The datum records inner slack only; a strictly inner-valued transport $\alpha\colon \mathcal{C} \to B(\mathrm{Inn}K)$ is already a bona fide functor to $B(\mathrm{Inn}K)$, so the $k_{g,f}$ are not what make α functorial – they record *which* inner automorphism witnesses the up-to-inner functoriality. We use throughout the inner/outer and centre sequences of K, which are precisely the two extensions (1); note that $\mathrm{Inn}K$ is at once the quotient $K/Z(K)$ of (II) and the kernel of (I), so the two extensions genuinely stack.

Cohomology over $N(\mathcal{C})$ is simplicial cohomology of the nerve; for a group Q of constant coefficients, a class in nonabelian $H^1(N(\mathcal{C}); Q)$ is a functor $\mathcal{C} \to BQ$ modulo object-relabelling, pointed by the trivial functor [4]. When $\mathcal{C} = B\Gamma$ for a group Γ, $N(\mathcal{C}) = B\Gamma$ computes ordinary group cohomology of Γ; the two bases we need are $\mathcal{C} = B\mathbb{Z}$ ($N \simeq S^1$) and $\mathcal{C} = B\mathbb{Z}^2$ ($N \simeq T^2$).

The band. Projecting the transport by $q\colon \mathrm{Aut}K \to \mathrm{Out}K$ kills the up-to-inner slack and yields a strict functor $\rho := q \circ \alpha\colon \mathcal{C} \to B(\mathrm{Out}K)$.

Definition 2. The *band* of the inventory gerbe is $\rho = q \circ \alpha$, with class $\beta := [\rho] \in H^1(N(\mathcal{C}); \mathrm{Out}K)$ (the Giraud stratum-two invariant). The band is *trivial* if $\beta = 0$.

That ρ is a genuine cocycle regardless of the inner slack is immediate: if $\alpha_{gf} = \iota\, \alpha_g \alpha_f$ with $\iota \in \mathrm{Inn}K = \ker q$, then $\rho(gf) = q(\alpha_{gf}) = \rho(g)\rho(f)$ because $q(\iota) = e$.

The inventory class. Once the band is trivial, the companion note constructs the stratum-three datum. Choose section-lifts $T_f \in K$ of the inner transport $\bar{\rho}\colon \mathcal{C} \to B(\mathrm{Inn}K)$; the defect $T_g T_f = \mathfrak{g}(g, f) \cdot T_{gf}$ lands in $\ker(K \to \mathrm{Inn}K) = Z(K)$ because both sides agree in $\mathrm{Inn}K$. Associativity in K, with $Z(K)$ central, makes $\mathfrak{g}$ a normalized 2-cocycle, and changing the section changes it by a coboundary.

Definition 3. The *inventory class* is $[\mathfrak{g}] \in H^2(N(\mathcal{C}); Z(K))$, the pullback along the inner transport of the class of the central extension (II). Its coefficient group is $Z(K)$, with trivial action (inner automorphisms fix the centre pointwise).

The floor. The two strata assemble into a single necessary-and-sufficient criterion, proved in the companion band-boundary note. First we fix *what* is to be globalized, retaining the gerbe's 2-cell datum rather than only its outer shadow.

Definition 4 (global inventory, D2)**.** A *global inventory* is a pair (μ, T) in which $\mu\colon \mathrm{Ob}\mathcal{C} \to \mathrm{Aut}K$ is a gauge and $T\colon \mathcal{C} \to BK$ is a *strict* functor, *retaining* the gerbe's 2-cell datum $k_{g,f} \in K$ (so that $\alpha_{gf} = \mathrm{conj}(k_{g,f}) \cdot \alpha_g \alpha_f$), subject to

$$\mu(y) \cdot \alpha(f) \cdot \mu(x)^{-1} = \mathrm{conj}(Tf) \qquad (f\colon x \to y).$$

A *strict global inventory* is such a pair. It is the full stratum-three globalization, not merely the bare outer datum.

Corollary 5 (the floor)**.** *Under Definition 4, a strict global inventory exists if and only if both strata vanish:* (i) $\beta = 0 \in H^1(N(\mathcal{C}); \mathrm{Out}K)$, *and* (ii) $[\mathfrak{g}] = 0 \in H^2(N(\mathcal{C}); Z(K))$, *where* (ii) *is well-defined once* (i) *holds.*

Proof sketch. A strict global inventory (μ, T) in the sense of Definition 4 has its gauge μ projecting by q to a trivialization of ρ, so (i) is necessary. Given (i), one may choose α inner-valued (band-triviality is exactly "inner-valued up to an outer relabelling"), whereupon – using the retained 2-cell datum $k_{g,f}$ – the gerbe is the honest gerbe of lifts of $\bar{\rho}$ through (II), neutral iff $[\mathfrak{g}] = 0$ by the central-extension lifting theorem [2, IV.3]. Full details, including the fungibility corollary (for an n-element commodity fibre $\mathrm{Out}(S_n) = 1$ for every n except $n = 6$, and $Z(S_n) = 1$ for $n \geq 3$ while S_2 is abelian so $[\mathfrak{g}] = 0$ there as well [5, Thms 7.5–7.7]; hence both strata vanish for *all* n and the only fungible exception is the band-only $n = 6$, $\mathrm{Out}(S_6) = \mathbb{Z}/2$), are in the band-boundary note [commit `e306ae5`]. □

Remark 6 (D1 versus D2)**.** Under the naive definition (D1) that drops the retained 2-cell datum $k_{g,f}$, the equivalence *fails*: the secondary stage is then a class in $H^1(\mathcal{C}; \mathrm{Inn}G)$ modulo $H^0(\mathcal{C}; \mathrm{Out}G)$, not in H^2. Rick's counterexample is $\mathcal{C} = B\mathbb{Z}$ with $G = S_3$ and $\alpha = \mathrm{conj}((01))$: there $\beta = 0$ and $[\mathfrak{g}] = 0$, yet no gauge μ strictifies the transport (Rick, band-boundary review, 2026-10-12). Retaining $k_{g,f}$, as in Definition 4, is exactly what restores the clean H^2 floor.

The floor says the two obstructions *stack*: clear the band, then clear the class. The rest of the paper asks whether clearing them is independent – and finds that on T^2 it is not.

3 The δ-commutator lemma

Fix an extension of groups

$$1 \longrightarrow M \xrightarrow{\iota} E \xrightarrow{\pi} Q \longrightarrow 1 \tag{2}$$

with M *abelian*. Then M is a Q-module by conjugation (π-equivariantly, the action descending because M is abelian), and a band $\rho\colon \mathbb{Z}^2 \to Q$ makes M a $\mathbb{Z}^2$-module: writing $\sigma := \rho(s)\cdot(-)$, $\tau := \rho(t)\cdot(-) \in \mathrm{Aut}M$ (commuting, as $\mathbb{Z}^2$ is abelian). Because $T^2 = B\mathbb{Z}^2$ is a $K(\mathbb{Z}^2,1)$ and M is abelian, $H^\bullet(T^2;M) = H^\bullet(\mathbb{Z}^2;M)$ is ordinary group cohomology. We will apply this with (E,Q,M) equal to $(\mathrm{Aut}K, \mathrm{Out}K, \mathrm{Inn}K)$ – the band-strictification extension (I), which needs K of nilpotency class ≤ 2 so that $\mathrm{Inn}K$ is abelian – and to $(K, \mathrm{Inn}K, Z(K))$ – the central extension (II), where $M = Z(K)$ is abelian with trivial action for every K.

3.1 The target group and the comparison map

Let $R = \mathbb{Z}[\mathbb{Z}^2] = \mathbb{Z}[s^{\pm1}, t^{\pm1}]$. The Koszul resolution of the trivial module $\mathbb{Z}$,

$$0 \to R\langle e_{st}\rangle \xrightarrow{\partial_2^K} R\langle e_s, e_t\rangle \xrightarrow{\partial_1^K} R\langle 1\rangle \to \mathbb{Z} \to 0,$$

with $\partial_1^K e_s = s-1$, $\partial_1^K e_t = t-1$, and $\partial_2^K e_{st} = (s-1)e_t - (t-1)e_s$, computes, after $\mathrm{Hom}_R(-,M)$, the complex $M \xrightarrow{d^1} M^2 \xrightarrow{d^2} M$ with $d^2(m_1,m_2) = (\sigma-1)m_2 - (\tau-1)m_1$, whence

$$H^2(\mathbb{Z}^2;M) = \mathrm{coker}d^2 = M\big/\big((\sigma-1)M + (\tau-1)M\big) = M_{\mathbb{Z}^2}, \tag{3}$$

the coinvariants. To read a bar cocycle in these coordinates we need the comparison map to the bar resolution $B_\bullet$ (free on normalized tuples $[g_1|\cdots|g_n]$, with $\partial_2^B[g|h] = g[h] - [gh] + [g]$).

Lemma 7. *The R-linear maps $\varphi_0(1) = [\,]$, $\varphi_1(e_s) = [s]$, $\varphi_1(e_t) = [t]$, and*

$$\varphi_2(e_{st}) = [s\,|\,t] - [t\,|\,s]$$

form a chain map $K_\bullet \to B_\bullet$ lifting $\mathrm{id}_{\mathbb{Z}}$. Consequently, for any bar 2-cocycle $f\colon (\mathbb{Z}^2)^2 \to M$,

$$\varphi^*[f] = [\,f(s,t) - f(t,s)\,] \in M_{\mathbb{Z}^2}. \tag{4}$$

Proof. Only the top degree needs checking. On one side $\varphi_1\partial_2^K e_{st} = (s-1)[t] - (t-1)[s]$. On the other,

$$\partial_2^B\big([s|t] - [t|s]\big) = \big(s[t] - [st] + [s]\big) - \big(t[s] - [ts] + [t]\big) = (s-1)[t] - (t-1)[s],$$

the terms $-[st] + [ts]$ cancelling because $st = ts$ in $\mathbb{Z}^2$. As a chain map between free resolutions of $\mathbb{Z}$ lifting the identity, φ induces the canonical isomorphism $\varphi^*\colon H^2_{\mathrm{bar}}(\mathbb{Z}^2;M) \xrightarrow{\sim} M_{\mathbb{Z}^2}$, unique up to homotopy, and (4) is its value on $[f] = [f\circ\varphi_2]$. □

Equation (4) is the precise form of "evaluate the obstruction cocycle on the fundamental 2-cell": the cell e_{st} maps to the antisymmetric expression $[s|t] - [t|s]$, the algebraic shadow of the torus relator $sts^{-1}t^{-1}$.

3.2 The lemma

Theorem 8 (δ-commutator lemma)**.** *With the extension* (2) *and band* $\rho\colon \mathbb{Z}^2 \to Q$ *as above, let* $A, B \in E$ *be any lifts of* $\rho(s), \rho(t)$*. Then* $w := [A,B] = ABA^{-1}B^{-1} \in \ker\pi = M$*, and*

$$\delta(\rho) = [\,w\,] \in M_{\mathbb{Z}^2} = H^2(T^2; M).$$

The class $[w]$ *is independent of the choice of lifts, and*

$$\delta(\rho) = 0 \iff \exists \text{ commuting lifts } A, B \iff \rho \text{ lifts to a homomorphism } \mathbb{Z}^2 \to E.$$

Here δ is the connecting map of the pointed sequence $H^1(T^2;E) \xrightarrow{\pi_*} H^1(T^2;Q) \xrightarrow{\delta} H^2(T^2;M)$; concretely, $\delta(\rho)$ is the class of the pullback extension $E_\rho = E \times_Q \mathbb{Z}^2 = \{(a,\gamma) : \pi(a) = \rho(\gamma)\}$ of $\mathbb{Z}^2$ by M, which splits iff ρ lifts to a homomorphism [2, IV.3–4]. Thus the vanishing criterion is classical; the content is the *value* $[w]$.

Proof. (i) $\pi(w) = [\rho(s), \rho(t)] = 1$ because $\rho(s), \rho(t)$ are images of the commuting generators s, t under a homomorphism; so $w \in \ker\pi = M$.

(ii) Work in E_ρ and choose lifts $\hat{s} = (A, s)$, $\hat{t} = (B, t)$; then $[\hat{s}, \hat{t}] = ([A,B], 1) = (w, 1)$, the commutator of the two lifts inside $\ker p \cong M$, where $p(a,\gamma) = \gamma$. Define the set-section $u(s^i t^j) := \hat{s}^{\,i}\hat{t}^{\,j}$ (every element of $\mathbb{Z}^2$ is uniquely $s^i t^j$; $u(1) = 1$, $p \circ u = \mathrm{id}$). Its factor set $f(g,h) := u(g)u(h)u(gh)^{-1} \in M$ is a normalized 2-cocycle representing $[E_\rho]$ [2, IV.3]. Using $ts = st$, so $u(ts) = u(st) = \hat{s}\hat{t}$,

$$f(s,t) = \hat{s}\hat{t}(\hat{s}\hat{t})^{-1} = 1, \qquad f(t,s) = \hat{t}\hat{s}(\hat{s}\hat{t})^{-1} = \hat{t}\hat{s}\hat{t}^{-1}\hat{s}^{-1} = [\hat{s},\hat{t}]^{-1} = w^{-1}.$$

Additively $f(s,t) = 0$, $f(t,s) = -w$, so by (4), $\delta(\rho) = \varphi^*[f] = [0 - (-w)] = [w] \in M_{\mathbb{Z}^2}$.

(iii) Lift-independence. Any other lifts are $A' = Au$, $B' = Bv$ with $u, v \in M$ (two lifts of the same class differ by $\ker\pi = M$). Moving every M-letter to the left in $[Au, Bv]$ – each picking up the action of the automorphisms to its left – gives, additively,

$$[Au, Bv] - [A,B] = \tau(\sigma - 1)v - \sigma(\tau - 1)u. \tag{5}$$

Since σ, τ commute with $\sigma - 1$ and $\tau - 1$, the right-hand side lies in $(\sigma-1)M + (\tau-1)M$; hence $[w]$ is well-defined in the coinvariants (3).

(iv) Vanishing. By the extension dictionary, $\delta(\rho) = 0$ iff E_ρ splits iff ρ lifts to a homomorphism; and by (5), commuting lifts exist iff some (u, v) makes $[Au, Bv] = 0$ iff $[w] = 0$ in $M_{\mathbb{Z}^2}$. □

Remark 9 (criterion versus slogan)**.** The vanishing criterion is "no commuting lifts", i.e. $w \notin (\sigma-1)M + (\tau-1)M$ – strictly weaker than $w \neq 0$ in M when the action is nontrivial. When $\sigma = \tau = \mathrm{id}$ (trivial action, as for every witness in §4–6) the coboundaries vanish, $M_{\mathbb{Z}^2} = M$, and the criterion is the clean $w = [A,B] \neq 0$. The identity (5) was additionally verified element-wise in E across a battery of class-two groups, including *nontrivial-action* pairs that no single trivial-action witness reaches.

3.3 The coupling criterion

Say K *couples on* T^2 *via* (2) if some band $\rho\colon \mathbb{Z}^2 \to Q$ has $\delta(\rho) \neq 0$ (then automatically $\beta = [\rho] \neq 0$, since the trivial band has $w = [\mathrm{id}, \mathrm{id}] = 0$). Theorem 8 converts this into a finite search.

Corollary 10 (coupling criterion)**.** K *couples on* T^2 *via* (2) *iff there exist commuting* $\bar{A}, \bar{B} \in Q$ *whose lifts to* E *never commute. In particular coupling forces* (2) *to be* non-split*; non-splitting is necessary but not sufficient.*

Proof. The equivalence is Theorem 8(iv). If (2) splits by $\theta\colon Q \to E$, then $\theta \circ \rho$ is a homomorphic lift of every band, so $\delta \equiv 0$; contrapositively coupling forces non-split. Necessity is not sufficiency: a band is a *commuting pair* of nontrivial classes, and a rank-one image never couples (since $H^2(\mathbb{Z}; -) = 0$); one needs the extension class to restrict nontrivially to some commuting $\mathbb{Z}^2 \to Q$, which can fail even when the class is nonzero – see §6. □

Because $S^1 = B\mathbb{Z}$ has $H^2(S^1; -) = 0$, the connecting map vanishes identically there: every band over a one-dimensional *free* base is strictifiable, and the two Giraud strata stack freely. Among the free bases $B(\mathbb{Z}^n)$ the torus is therefore the first where coupling can occur, and Corollary 10 says exactly when it does. This "first" is relative to the free bases only. Over a *torsion* base coupling already appears at rank one: $B(\mathbb{Z}/2)$ has nerve $\simeq \mathbb{RP}^\infty$ with $H^2(B(\mathbb{Z}/2); -) \neq 0$, and $K = D_4 \times C_2$ with the outer involution $a\colon (r \mapsto r,\ s \mapsto sc,\ c \mapsto r^2c)$, $a^2 = \mathrm{conj}(g) \neq 1$, admits no strict lift (Rick, band-boundary review, 2026-10-12). So the torus is the first *free* base, not the first base overall.

4 A coupled witness on the torus

We exhibit a structure group that couples, via both extensions (1) at once. The following is *computed*: a finite brute-force certificate over the full (machine-verified complete) automorphism group, with the arithmetic core independently re-checked in Lean.

The group. Take $K = D_4 \times C_2 = \langle r, s\rangle \times \langle c\rangle$ with $r^4 = s^2 = c^2 = 1$, $srs^{-1} = r^{-1}$, c central. Then K has nilpotency class two,

$$Z(K) = \{1, r^2, c, r^2c\} \cong (\mathbb{Z}/2)^2, \quad \mathrm{Inn}K = K/Z(K) \cong (\mathbb{Z}/2)^2 = \langle [r], [s]\rangle, \quad |\mathrm{Out}K| = 16,$$

and $|\mathrm{Aut}K| = 64$ (verified). The second central involution c – absent from D_4 and Q_8 alone – is what lets automorphisms mix the three central involutions r^2, c, r^2c and so breaks the splitting of (I).

The band (strictification coupling, extension (I)). Take

$$A\colon\ r \mapsto r^3,\ s \mapsto s,\ c \mapsto r^2c, \qquad B\colon\ r \mapsto r^3c,\ s \mapsto s,\ c \mapsto c.$$

Both are automorphisms, both are *outer*, both have order two in $\mathrm{Out}K$ and commute, so $\rho\colon \mathbb{Z}^2 \twoheadrightarrow \langle [A], [B]\rangle \cong (\mathbb{Z}/2)^2 \hookrightarrow \mathrm{Out}K$, $s \mapsto [A]$, $t \mapsto [B]$, is a nontrivial band: $\beta \neq 0$. Both act trivially on $\mathrm{Inn}K = K/Z(K)$ (e.g. $A(r) = r^3 \equiv r$, $A(c) = r^2c \equiv c \bmod Z$), so the action is trivial and $H^2(T^2; \mathrm{Inn}K) = \mathrm{Inn}K = (\mathbb{Z}/2)^2$. The commutator of lifts is

$$[A, B] = c_s \text{ (conjugation by } s), \qquad \delta(\rho) = [A, B] = [s] \neq 0 \in H^2(T^2; \mathrm{Inn}K) = (\mathbb{Z}/2)^2.$$

Exhaustively over all 16×16 inner adjustments (Au, Bv), none commute, so by Corollary 10 the band is *not* strictifiable: the stratum-two band forces the degree-two strictification class. The strata do not split.

The central coupling (extension (II)). The same group couples the inventory class. The inner band $s \mapsto [r]$, $t \mapsto [s]$ lifts to K with commutator $[r, s] = r^2 \in Z(K)$, so $[\mathfrak{g}] = [r^2] \neq 0 \in H^2(T^2; Z(K)) = (\mathbb{Z}/2)^2$ (trivial action). Thus $D_4 \times C_2$ realizes the coupling at both stages of the tower (1): $\mathrm{Out}K \leftarrow \mathrm{Aut}K$ (coeff $\mathrm{Inn}K$) and $\mathrm{Inn}K \leftarrow K$ (coeff $Z(K)$).

Remark 11 (grade and scope)**.** Both couplings are finite brute-force certificates (*computed*); the arithmetic core – that A, B are commuting outer automorphisms acting trivially on $\mathrm{Inn}K$, that $[A, B] = c_s \neq 0$, and that no inner adjustment makes the lifts commute – is machine-checked in Lean (*finite core lean-verified*). We do *not* claim $D_4 \times C_2$ is minimal over all groups of order ≤ 16; only that it is the smallest coupler in the enumerated battery of §6.

5 The sharp line: the n-lot ring

The witness shows coupling *can* occur. The n-lot ring pins down *exactly when* the inventory class (extension (II)) is essentially nonzero, as an arithmetic condition on n. Here the result is *proved*, with the finite split-test core lean-verified.

The structure group. Let $F(n)$ be n fungible lots arranged in a ring – the vertices of the n-cycle C_n with cyclic adjacency, $n \geq 3$. Its relabelling group is the dihedral group

$$K(n) = \mathrm{Aut}(C_n) = D_n = \langle r, s \mid r^n = s^2 = 1,\ srs^{-1} = r^{-1} \rangle \quad (\text{order } 2n).$$

Call the inventory class $[\mathfrak{g}]$ *essentially nonzero* if some base $(\mathcal{C}, \bar{\rho})$ with inner transport gives $[\mathfrak{g}] \neq 0$ – i.e. strict global inventory is obstructed for some supply chain on this fibre.

Theorem 12 (sharp n-lot criterion)**.** *For $K = D_n$,*

$$[\mathfrak{g}] \text{ essentially nonzero} \iff \text{the central extension (II) is non-split} \iff 4 \mid n.$$

Equivalently, strict global inventory on the n-lot ring is always rectifiable unless $4 \mid n$.

Proof. Centre and inner group. A rotation r^k is central iff $r^{2k} = 1$, giving $k = 0$ (all n) and $k = n/2$ (n even); a reflection is never central for $n \geq 3$. Hence $Z(D_n) = 1$ for n odd and $\langle r^{n/2} \rangle \cong \mathbb{Z}/2$ for n even, and $\mathrm{Inn}D_n = D_n/Z(D_n)$ is D_n (n odd) or $D_{n/2}$ (n even). For n odd, $Z = 1$, so (II) is $1 \to 1 \to D_n \to D_n \to 1$ and the coefficient group $Z(D_n) = 1$ makes $[\mathfrak{g}]$ *vacuously* zero; the question lives in the even case.

The split jump. Fix $n = 2m$, $m \geq 2$. A splitting of $1 \to \langle r^m \rangle \to D_{2m} \to D_m \to 1$ is a complement H to the central $Z = \langle r^m \rangle$; being of index two it is normal, so a complement makes $D_{2m} = Z \times H$ an internal direct product with $H \cong D_m$. Thus

$$\text{(II) splits} \iff D_{2m} \cong \mathbb{Z}/2 \times D_m.$$

Compare centres, an isomorphism invariant: $|Z(D_{2m})| = 2$ always, while $|Z(\mathbb{Z}/2 \times D_m)| = 2\,|Z(D_m)|$ equals 2 iff $|Z(D_m)| = 1$ iff m is odd. So for m even the two groups are non-isomorphic and (II) is non-split; for m odd an explicit complement exists, namely $H = \langle r^2, s \rangle \cong D_m$ (since m odd makes r^2 of order m and $r^m \notin \langle r^2 \rangle$, so $H \cap Z = 1$ and $|HZ| = 4m = |D_{2m}|$). Hence (II) is non-split iff n even and $n/2$ even, i.e. iff $4 \mid n$.

From splitting to the class. With trivial coefficients, $[\star] = 0$ in $H^2(\mathrm{Inn}K; Z(K))$ iff (II) splits ([2, IV.3]; [3]). Taking $\mathcal{C} = B(\mathrm{Inn}K)$, $\bar{\rho} = \mathrm{id}$ realizes $[\mathfrak{g}] = [\star]$, while for any $(\mathcal{C}, \bar{\rho})$, $[\mathfrak{g}] = \bar{\rho}^*[\star] = 0$ whenever $[\star] = 0$. So $[\mathfrak{g}]$ is essentially nonzero iff $[\star] \neq 0$ iff (II) non-split iff $4 \mid n$. □

Remark 13 (the D_6 reconciliation)**.** The six-lot ring is the witness that "nonabelian with nontrivial centre" is *necessary but not sufficient* for $[\mathfrak{g}] \neq 0$. D_6 is nonabelian with $Z(D_6) = \langle r^3 \rangle \cong \mathbb{Z}/2 \neq 1$, so it satisfies the naive hypothesis; but $6 = 2 \cdot 3$ with $m = 3$ odd, so by the split jump $D_6 \cong \mathbb{Z}/2 \times S_3$ (complement $\langle r^2, s \rangle \cong S_3$), extension (II) splits, and $[\mathfrak{g}] = 0$ on every base. The honest dividing line is non-splitting of (II), which on the n-lot ring is precisely $4 \mid n$ (yes for $n = 4, 8, 12, 16, \ldots$; no for $n = 6, 10, 14, \ldots$ and all odd n).

The split test was run for $n = 3, \ldots, 16$ and the arithmetic core – the centre table, the explicit complement for $4 \nmid n$, the centre-mismatch obstruction for $4 \mid n$, and the $D_6 \cong \mathbb{Z}/2 \times S_3$ reconciliation – is machine-checked in Lean, axiom-free (no `propext`, `sorryAx`, or `native_decide`).

6 Coupling is rare

Corollary 10 makes coupling a finite search: enumerate the commuting pairs in $\mathrm{Out}K$ and test whether any has no commuting lift in $\mathrm{Aut}K$. We ran this over a battery of nilpotency-class-two groups built from explicit multiplication tables, computing Aut, Inn, Out by brute force and cross-checking $|\mathrm{Aut}K|$ against known values to confirm the engine returns the *complete* automorphism group (a missed automorphism could fake a "no commuting lift"). This is *computed*, and over the listed family only – not a census of all groups of a given order.

K	$\lvert\mathrm{Aut}K\rvert$	$\lvert\mathrm{Inn}K\rvert$	$\lvert\mathrm{Out}K\rvert$	couples via (I)?
$D_4,\ Q_8$	8, 24	4	2, 6	no (split)
$\mathrm{Heis}(\mathbb{F}_3)$	432	9	48	no (split)
$M_{16},\ C_4 \rtimes C_4$	16, 32	4	4, 8	no
$\mathrm{ES}^+_{32} = 2^{1+4}_+$	1152	16	72	no (split)
$\boxed{D_4 \times C_2}$	64	4	16	**yes (30 pairs)**
$Q_8 \times C_2$	192	4	48	yes (54 pairs)

Only $D_4 \times C_2$ and $Q_8 \times C_2$ couple; every group whose extension (I) splits decouples, as Corollary 10 demands. The instructive rows are M_{16} and $C_4 \rtimes C_4$: their extension (I) need not split, yet they carry *no* coupled band. Non-splitting is genuinely weaker than coupling – exactly the gap flagged in Corollary 10: one further needs a *commuting* outer pair whose lifts cannot be made to commute. Coupling therefore requires two independent conditions to hold at once: a non-split automorphism tower *and* two commuting outer involutions with no commuting lifts. The richest composition failures live in this thin slice.

7 Consequences

For a torus-shaped supply chain – two independent, commuting, reversible operations – the two stages of the nonabelian inventory obstruction cease to be independent. The band-to-strictification connecting map is given exactly by a commutator of automorphism lifts (Theorem 8), so a nonzero band *forces* a nonzero degree-two class. The same engine, applied to the central extension, shows the inventory class itself couples; and it locates both phenomena sharply (the n-lot ring fails exactly when $4 \mid n$, Theorem 12) and rarely (a handful of structure groups across an enumerated family, §6).

For the grant's Applications pillar the reading is concrete. Global stock-keeping on a cyclic, reconvergent logistics network can fail *compositionally*, with no local remedy, but only when the stock fibre carries genuine gauge structure – a relabelling group with non-split automorphism tower – and only on a base rich enough to hold a degree-two class. Plain fungibility (symmetric-group fibres) never triggers it; the economically meaningful regime is allocation defined up to a gauge group. That the dangerous structure groups form a thin, explicitly identifiable slice turns a potential liability into a design criterion: one can read off, from the automorphism tower of the stock type alone, whether a two-cycle in the network can obstruct reconciliation.

Open edges. Three are worth naming, each a clean next target rather than a gap in what is claimed here. *(1) Minimality.* The rarity enumeration covers a listed family, not a census; whether $D_4 \times C_2$ is the unique minimal coupler over *all* groups of order ≤ 16 is open. *(2) Higher genus.* The δ-commutator lemma is special to $\mathbb{Z}^2$ only through the two-generator one-relator Koszul resolution; a genus-g base would replace the single relator by a product of commutators, and the class by a sum of commutators of lifts – a natural extension of §3. *(3) Nonabelian kernel.* The class-two hypothesis makes $\mathrm{Inn}K$ abelian, so δ lands in ordinary H^2; dropping it replaces the clean connecting map by an Eilenberg–Mac Lane obstruction valued in a torsor of centres, genuinely harder and not needed for the inventory lane.

References

[1] L. Breen, *Bitorseurs et cohomologie non abélienne*, in: The Grothendieck Festschrift, Vol. I, Progr. Math. 86, Birkhäuser, 1990, pp. 401–476.

[2] K. S. Brown, *Cohomology of Groups*, Graduate Texts in Mathematics 87, Springer, 1982 (Chapter IV).

[3] S. Eilenberg and S. Mac Lane, *Cohomology theory in abstract groups. II. Group extensions with a non-Abelian kernel*, Ann. of Math. (2) **48** (1947), 326–341.

[4] J. Giraud, *Cohomologie non abélienne*, Grundlehren der mathematischen Wissenschaften 179, Springer, 1971 (Chapters III–IV).

[5] J. J. Rotman, *An Introduction to the Theory of Groups*, 4th ed., Graduate Texts in Mathematics 148, Springer, 1995 (Theorems 7.5–7.7).